Lab 3: Component Modelling & Architectural Pattern Selection

Name: Anusha G
Sec:B
SRN:PES2UG24CS075
Sl.No 12

Blood Bank Inventory & Emergency Donor Matcher

LAB 3 — ARCHITECTURE SELECTION AND COMPONENT JUSTIFICATION

System: Blood Bank Inventory & Emergency Donor Matcher

Architecture Selection

Layered Architecture for the Blood Bank Inventory & Emergency Donor Matcher system.

Architectural Choice

The system is organized into separate layers/components responsible for handling emergency requests, blood inventory, donor management, donor matching, and notifications. This separation provides a clear structure between user requests, business processing, and data management.

Reason 1 — Maintainability

The Blood Bank system contains different functions such as inventory management, donor management, emergency request processing, and donor matching. Layered architecture separates these responsibilities, making the system easier to develop, test, maintain, and modify. For example, changes to the donor-matching logic can be made without significantly affecting the inventory management component.

Reason 2 — Emergency Response

Emergency blood requests require quick processing. The architecture separates the matching and inventory functions into dedicated components, allowing the system to process blood availability and donor-matching operations efficiently. This makes the architecture suitable for handling urgent requests.

Security Advantage

Layered architecture allows security controls to be applied at different levels. Authentication and authorization can be implemented at the application layer, while sensitive donor and patient information can be protected at the data-access and database levels. Components can also communicate through controlled APIs rather than directly exposing database information.

Performance Benefit

The separation of components allows frequently used operations such as blood availability checking and donor matching to be optimized independently. Matching requests can access only the required donor and inventory information, reducing unnecessary processing and improving response time during emergencies.

Conclusion

Layered architecture provides a simple, maintainable, secure, and efficient structure for the Blood Bank Inventory & Emergency Donor Matcher system. The identified components and interfaces clearly define how emergency requests, blood inventory, donor information, matching operations, and notifications interact.

